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Research paper

Computational study of the ring opening mechanism of substituted temolozolomide, TMZ-R (R = Cl, OH, CF_3) in water solvent

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HIGHLIGHTS

- Pathway-1 and pathway-2 occur kinetically and thermodynamically.
- Substitution tends to increase the energy barrier.
- The lowest conformer of MTIC acid is proposed.
- IHBs are disrupted in MTIC acids to form solute-solvent IHBs.

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ABSTRACT

MP2/6-311 + G(d,p) was employed to study the mechanism of the initial step of ring opening of temozolomide (TMZ-H) and its substituted, TMZ-R (R = Cl, OH, CF3) analogues. The study reveals that the difference formation energies of substituted MTIC acids with respect to MTIC acid are ranging between 0.44 and 1.58 kcal/mol. The ring opening of TMZ-R along pathway-2 leads to the lowest conformer (D), whereas reactions along pathway-1 give rise to lower transition states. The barrier energies of TMZ-R are 1.60–3.75 kcal/mol larger than TMZ-H. The conformational analysis reveals that intramolecular hydrogen bonds do not have a stabilizing effect in water solvent.

1. Introduction

Temozolomide (TMZ-H) is an antitumor prodrug belonging to the imidazotetrazine molecular class of bicyclic aromatic heterocycles. As an antitumor prodrug [1,2], TMZ-H continues to attract the attention of numerous research groups owing to its unique activity against high-grade glioma. The activation mechanism for TMZ-H has been elucidated by means of NMR spectroscopy; it starts by ring opening of the initial structure [3,4]. TMZ-H tetrazinone ring opening, which eventually leads to the complete degradation of the initial system, is very clean and is a base catalyzed hydrolysis through nucleophilic attack at the carbon atom of the carbonyl. This ring opening reaction begins with the addition of a water molecule to the carbonyl moiety. Subsequently, by CO_2 elimination, the bioactive intermediate molecule, which is methyltriazen-1-yl-imidazole-4-carboxamide (MTIC), is generated. Thereafter, the MTIC decomposes and releases the highly reactive methylating species, methyldiazonium ion, and the 5-amino-1H-imidazole-4-carboxamide product. The 2D pathways of the hydrolytic degradation mechanism of TMZ-H to MTIC were illustrated by Mirzaei et al. [5] (Scheme 1).

The methyldiazonium cation, an active methylating agent, is vulnerable to instantaneous nucleophilic attack by electron donors within DNA nucleotides, causing the transfer of a methyl group from the ion to form an O6-methylguanine adduct which confers the pharmacological activity. This adduct can be repaired by O6-methylguanine-DNA methyltransferase (MGMT), resulting in active resistance to TMZ-H in tumors in which MGMT is proficient [6]. It is worthwhile to mention that the rate of TMZ-H degradation is highly pH sensitive [5,7]. The half-life of TMZ-H in phosphate buffer at physiological pH (7.4) is around 1.83 h whereas the half-life in acidified human plasma (pH < 4) increases to about 24 h [7]. Although the C6 atom is established as the reactivity indicator, there is a dearth of current information regarding how substituents affects the magnitude of the partial charge on the C6 atom and hence the propensity to initiate the hydrolysis reaction.

Early history of the activity of various TMZ-H derivatives, which are, N7-substituted against TLX5 lymphoma implanted s.c. in CBA/CA

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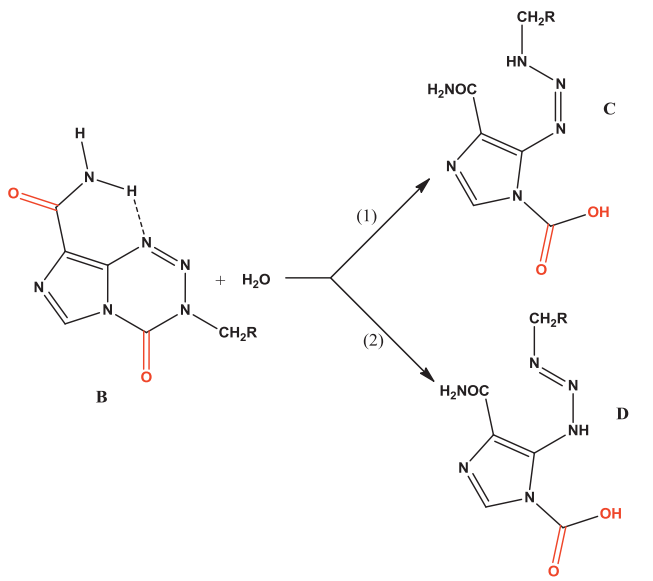


Scheme 1. Different pathways for the direct hydrolysis of substituted Temozolomide.

mice, indicated that the 7-haloethyl analogues had a better activity than TMZ-H, the 7-methyl was of moderate effectiveness but other analogues, as exemplified by the 7-ethyl, were inactive [8]. Moreover Wheelhouse et al. [4] have observed by NMR that when the N7-substituent is CF_3, the 7-trifluoroethyl analogue (TMZ-CF_3) behaves differently. It is much less stable than TMZ-H in buffer or DMSO and appears to ring open by retrocycloaddition to diazoimidazole carboxamide and trifluoroethyl isocyanate. In this way, as 7-substituent becomes more electron-withdrawing, the rate of degradation of the tetrazine increases. A correlation between the ^{1}H chemical shift of the CH_2 adjacent to N7 and the hydrolysis rate was found in using the following 7-substituents: N7-CH_2-H (TMZ-H), N7-CH_2-CH_2-NH_2 (TMZ-CH_2NH_2), N7-CH_2CH_2-O-alkyl (TMZ-CH_2-O-alkyl) and N7-CH_2CH_2-OPh (TMZ-CH_2-O-Ph) [9]. If an electron-withdrawing N7-substituent increases the rate of hydrolysis, the addition of an electron-donating group at the equivalent position on the other side of the C6 = O11, i.e C4 should slow the reaction back down again. Thus, more mechanistic details are needed to explain this structure-reactivity correlation.

Furthermore, despite experimental instances on ring opening in various conditions, the fundamental basis on the energetics of this hydrolysis-induced ring opening is still unclear and remains poorly understood. For instance a little is known on the energetic barriers of the transition and the global minimum of the ring-opened species obtained after the first step of hydrolysis of TMZ-H. The nature of the optimized geometry of the ring opened species suggested in Ref. [5] is questionable since the potential energy surface of this species was not investigated in detail. Herein, the present study employs MP2 theory to investigate how the electron-withdrawing N7-substituents affect the mechanism of the initial step of ring opening (Scheme 1) and enhance the propensity to initiate the hydrolysis reaction. Although it seems that the energy-favored overall mechanism is a combination of the two paths (tautomerization may occur during the process) *in vivo* [5]. The energetic barriers of the transition states and the global minimum of the ring-opened species (**C** and **D**) are also examined. The electron-withdrawing N7-substituents considered in this study are $-CH_2OH$, $-CH_2CF_3$, and $-CH_2Cl$. These substituents were chosen, by taking into account their empirical Taft constants [10]. The last being adapted to reactivity problems for compounds bearing bulky substituent close to the reaction center.

It is known that TMZ-H is widely used for brain cancer treatment but it is activity is quite limited. It is become more important to look for new molecule with enhanced bioactivity. Therefore, we investigated theoretically substituted TMZ-R which have been reported experimentally to be more active than TMZ-H. As far as we know, no theoretical studies were performed to elucidate the ring opening reaction of TMZ-R induced by water. This work attempts to predict the effect of R substituents on the ring opening of TMZ-R upon water attack using high accurate quantum chemistry methods and investigate the relative stabilities of different conformers of the ring open species, namely, the 5-(3-methyltriazen-1)imidazole-4-carboxamide acid denoted as MTIC-acid.

The initial step of the ring opening of TMZ-H is considered to be activated by a water attack to C6=O11 double bond, leading to the opening of the six-membered ring in TMZ-H. We now report a computational study of the ring opening of TMZ-R based on the mechanism of TMZ-H (Scheme 1) as the simplest model. This study also provides the intramolecular hydrogen bonds in open ring conformers of substituted TMZ-R analogues and energy cost of the ring opening reaction of TMZ-R. The water-phase stationary points on the potential energy surface for ring-opening reaction of the TMZ-R molecules are established at the MP2/6-311 + G(d,p) level of theory. Transition states have been confirmed by frequency and intrinsic reaction coordinate (IRC) calculations.

The paper is organized as follows:

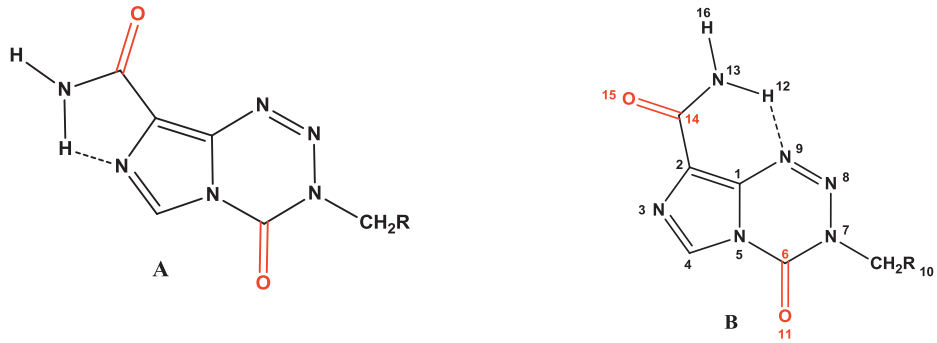
- Firstly, the effect on the substitution on TMZ-R is analyzed in water. The results show that the replacement of H by CF_3, Cl and OH can increase the stability of temozolomide in water
- Secondly, a comparison of the water interaction on C6=O in TMZ-R was analyzed. The interaction was found weak; the substitution in TMZ-H does not increase the interaction energy. Hence, the less bound water can jump over the ring to interact with C6=O to initiate the ring opening reaction.
- Thirdly, an analysis of the effect of Cl, CF_3 and OH substituent on the transition states and reaction energy of the ring opening of the six-member ring in TMZ-R in water solvent is performed using SMD solvation model and intrinsic reaction coordinate (IRC) method. The substituted MTIC acids molecules are stable based on the computed reaction energies but do not lower the barrier energies.
- Fourthly, the study also analyses the intramolecular hydrogen bonds in conformers of MTIC acids and their relative stabilities. The lowest conformer was identified and the intramolecular hydrogen bonds are suspected to disrupt in water in favor of solute-solvent intermolecular hydrogen bonds.

2. Computational methods

The geometries of substituted TMZ-H derivatives, denoted as TMZ-R (with R = OH, Cl, CF_3), were fully optimized using MP2/6-31 + G(d,p) [11] as implemented in the Gaussian 09 software package [12] in solvent employing the SMD solvation model [13]. Single point calculations from MP2 optimized geometries were performed at 6-311 + G(d,p) to get more refined relative energies. All minima reported in the present work were characterized by real vibrational frequencies and each transition state by one imaginary mode. To verify if transition state is connected to products, intrinsic reaction coordinate calculations (IRC) [14] were carried out at M062X/6-31G(d) owing to the time consuming of MP2 method. M062X functional [15] was reported to be one of the best and robust functional for organic systems [16]. Entropic effects were not taken into account in the evaluation of the activation energies.

TMZ-R can exist in two different rotamers A and B like TMZ-H. The A and B rotamers and the numeric labels of atoms in different TMZ-R structures used in the present study are illustrated in Scheme 2.

Two pathways for the direct hydrolysis of substituted temozolomide (Scheme 1) were carefully examined referring to TMZ-H hydrolysis [5]. The ring opening reaction illustrated in this Scheme begins with the addition of a water molecule to the C6=O11 carbonyl moiety [3,4]. It is worth to point out that the equilibrium geometry of [TMZ-H…H_2O]



Scheme 2. Rotamers of substituted temozolomide (TMZ-R) and atom labels used in the present work.

complex considered in this study is that proposed by Mirzaei et al. [5] and differs to that of Kasende et al. [17–23], in which studies, C6═O11 was found less proton-acceptor than C14═O15. To get the global minimum of the open chain (MTIC acid) obtained after C6-N7 bond breaking upon water attack on C6-O11 site of TMZ-H (**C** and **D**), double scan rotation with an increment of 15° of two dihedral angles (C1-N9-N8-N7 and C1-N5-C6-O11) were performed at B3LYP/6-31 + G(d,p) [24] for the two open ring species through path1 and path2 and transition states were built. All conformers identified on the two potential energy surfaces were then reoptimized at MP2/6-31 + G(d,p) and relative energies computed at MP2/6-311 + G(d,p). GaussView 05 [25,26] program was used for visualization.

3. Results and discussion

3.1. Solvation energies in water of TMZ-R (R = H, CF_3, Cl, OH)

Table 1 lists the free energies of solvation, HOMO-LUMO gaps (H-L in eV) of TMZ-R, total electronic energies difference between rotamers (Ediff in kcal/mol) and NICS values of TMZ-R in water solvent computed at MP2/6-311 + + G(d,p) using SMD polarizable continuum model. The data listed in the Table shows that the substitution of H increases slightly the solute–solvent interaction. The computed solvation energies of TMZ-H, TMZ-CF_3, TMZ-Cl and TMZ-OH were estimated, respectively, nearly −28.95, −31.57, −29.46, and −34.72 kcal/mol. Thus, as one can expect the strongest solute-water interaction is attributed to TMZ-OH. Except, TMZ-OH for which rotamers have quite same relative stability in water (energy difference of 0.36 kcal/mol), the relative energies between rotamers of TMZ-H, TMZ-Cl and TMZ-CH_3 are around 0.7 kcal/mol. This result suggests rotamers of type **A** to be slightly more stable than **B** (Scheme 2). The HOMO-LUMO gaps of these TMZ-R are also similar (H-L = 9.82–9.91 eV). HOMO-LUMO gap is an important hardness index widely used in the evaluation of stability of compounds. However, on closer inspection, substitution of H by R groups (e.g. Cl, OH and CF_3) increases slightly the HOMO-LUMO gaps by ~0.1 eV indicative of a slight increase in stability. TMZ-R consists of five- and six-membered rings fused together. The nucleus independent chemical shift (NICS) values are significantly higher at the middle of the five member ring than six-member ring illustrating its aromatic character and stability.

Table 1
Solvation energies (Gsolv in kcal/mol), non-electrostatic contribution (non Gel in kcal/mol), HOMO-LUMO gaps (H-L in eV) of TMZ-R, total electronic energies difference between rotamers (Ediff in kcal/mol) and NICS values at the middle of six member ring (five member ring in parenthesis) computed at MP2/6-311 + G(d,p) in water solvent.

R	△Gsolv	non Gel	H-L	E diff	NICS(0)
H	−28.95	4.16	9.82	0.64	−2.50 (−16.41)
CF_3	−31.57	5.73	9.91	0.75	−1.91 (−16.23)
Cl	−29.46	4.02	9.89	0.79	−2.20 (−15.62)
OH	−34.72	4.20	9.91	0.36	–

3.2. Interaction between TMZ-R and water on C6═O11 site

The binding energies of the various complexes formed by water attack on C6═O of TMZ-R as well as the optimized H-bonds and NBO charge on C6 atom are collected in Table 2. The interaction between water and carbonyl group to initiate ring opening is probably electrostatic. The atomic charge at C6 is suspected to play an important role in this ring opening. The NBO charges at C6 atom for different TMZ-R are about 0.82. The interaction between water and TMZ-R on C6═O site is weak as indicated by the values of water binding energies of about −10 kcal/mol, red-shift of about 30 cm^{-1} and H-bond lengths of about 0.2 Å (Table 2 and Fig. 1). However, the ring opening is initiated by water adding on C6═O carbonyl on six member ring of TMZ rather than on the C14═O15 on the five-member ring. This shows that the five-member ring of TMZ-R is more rigid to bond breaking than the six-member –ring.

3.3. Analysis of the ring opening of TMZ-R initiated by water

It is well known that search of transition states is of great importance in predicting reaction mechanisms. It was reported that the hydrolysis of TMZ-H can proceed through two different mechanisms illustrated in Scheme 1:

$$\mathrm{TMZ-R}(\boldsymbol{B}) + \mathrm{H_2O} \rightarrow [\mathrm{TMZ{-}R}\ldots\mathrm{H_2O}]^{\#}(\boldsymbol{TS}) \rightarrow \text{MTIC acid}\ (\boldsymbol{C})(\text{path - 1})$$

$$\mathrm{TMZ-R}(\boldsymbol{B}) + \mathrm{H_2O} \rightarrow [\mathrm{TMZ{-}R}\ldots\mathrm{H_2O}]^{\#}(\boldsymbol{TS}) \rightarrow \text{MTIC acid}\ (\boldsymbol{D})(\text{path - 2})$$

The same mechanisms are examined beneath for its counterparts TMZ-R compounds.

The imaginary vibrational frequencies and geometric features of transition states (**C**) related to the transformation of TMZ-R in MTIC acids along pathway-1 are given in Fig. 2 and Table 3. These geometrical features show clearly that the water molecule in TS dissociates in

Table 2
Binding energies (BE in kcal/mol), hydrogen bonds lengths (H-bonds in Å), red-shift of O—H bonds (△ν O-H in cm^{-1}) and NBO charge on C6 (Q in a.u) in TMZ-R…H_2O complexes calculated at MP2/6-311 + G(d,p). (a) Red shift frequencies (△ν) were computed at MP2/6-31 + G(d,p).

R	BE	C═O…H	△ν O-H(a)	Q
H	−10.66	1.977	30.72	0.814
CF_3	−10.09	1.979	28.43	0.826
Cl	−10.18	1.986	28.17	0.826
OH	−9.40	1.968	30.12	0.825

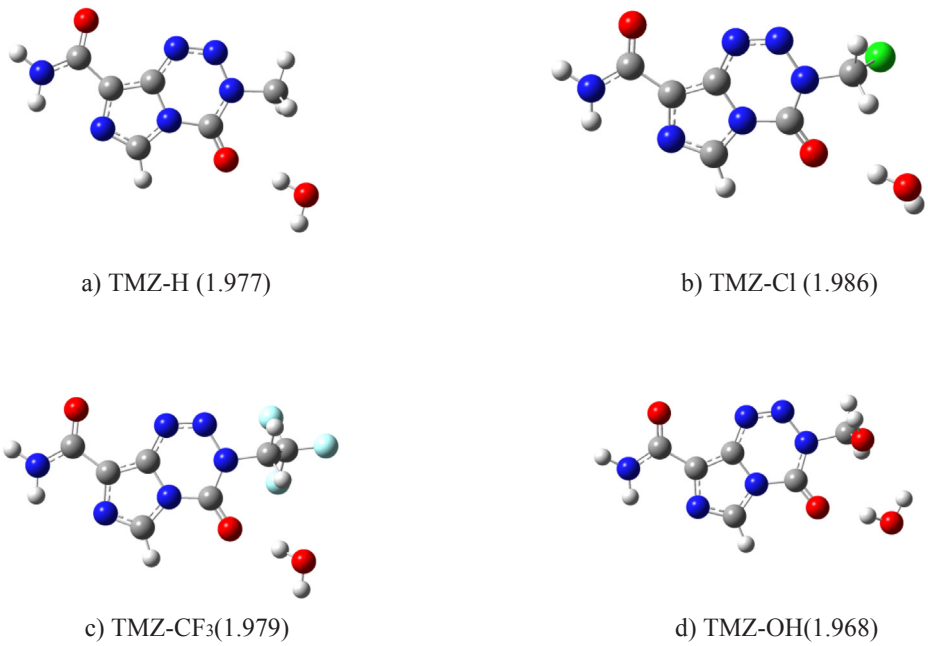


Fig. 1. TMZ-R…H_2O complexes and O11…H—O bonds distances in parenthesis optimized at MP2/6-31 + G(d,p) (cfr Scheme 2 for atomic labels).

OH and H so they can bind to C6 and N7 atoms that results in the weakening of C6-N7 bonds. The C6-N7 bonds in those transition states (TS) are 0.004–0.066 Å longer than TS of TMZ-H. The N7…H—O bonds are ranging in between 1.287 and 1.291 Å. These transition states reported as saddle points have each one imaginary vibration mode located at *i1669* cm^{-1} for TMZ-H, *i1628* cm^{-1} for TMZ-CF_3, *i1621* cm^{-1} for TMZ-Cl and *i1672* cm^{-1} for TMZ-OH (Fig. 2). The activation energies of MTZ-CF_3, TMZ-Cl, and TMZ-OH are 1.60–3.75 kcal/mol larger than that of TMZ-H estimated about 54.21 kcal/mol at MP2/6-311 + G(d,p). The second path that leads to the ring opening of TMZ-R is accompanied by a proton transfer from water to N9 atom and breaking of C6—N7 bond, with **D** the highest point on the potential energy. The optimized geometries and activation energies of **D** along pathway-2 are shown in Fig. 3 and are characterized by following imaginary frequencies: *i1683* cm^{-1} for TMZ-H, *i155* cm^{-1} for TMZ-CF_3, *i157* cm^{-1} for TMZ-Cl and *i1732* cm^{-1} for TMZ-OH. Their geometric parameters are given in Table 3. The N9…H—O bonds in these

Table 3
Selected bonding distances in Å (N9…H—O, N7…H—O, O—H and C6—N7) of different substituted transition states along path-1 and path-2 in parenthesis computed at MP2/6-311 + G(d,p).

TS of TMZ-R	N9…H—O	N7…H—O	O—H bound to N	C6—N7
TMZ-H	1.333 (2.999)	2.263 (1.289)	1.242 (1.243)	1.607 (1.590)
TMZ-CF_3	1.363 (2.928)	2.283 (1.291)	1.221 (1.243)	2.364 (1.642)
TMZ-Cl	1.341 (2.932)	1.972 (1.289)	1.269 (1.240)	2.389 (1.656)
TMZ-OH	1.319 (3.022)	2.258 (1.287)	1.265 (1.252)	1.618 (1.594)

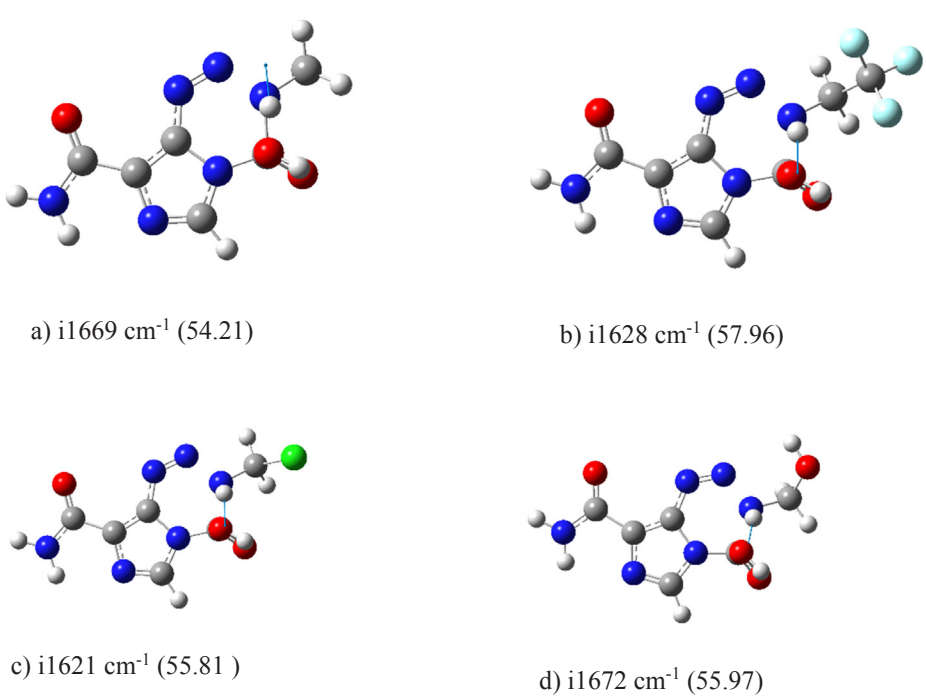


Fig. 2. Imaginary vibrational modes of the transition states (**C**) computed at MP2/6-31 + G(d,p) and their relative energies in kcal/mol computed at MP2/6-311 + G(d,p) in parenthesis with respect to TMZ-R + H_2O molecular systems for the formation of substituted MTIC acids along path-1. Where R stands for H (a), CF3 (b), Cl (c) and O-H (d).

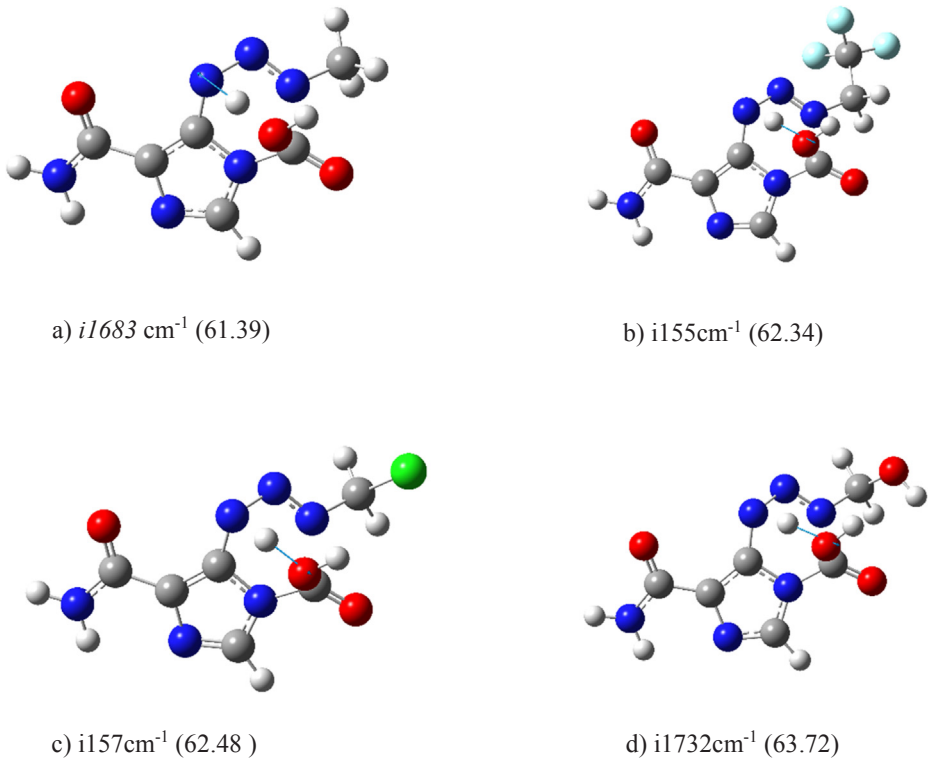


Fig. 3. Imaginary vibrational modes of the transition states (**D**) computed at MP2/6-31 + G(d,p) and their single point relative energies in kcal/mol computed at MP2/6-311 + G(d,p) in parenthesis with respect to TMZ-R + H_2O molecular systems for the formation of substituted MTIC acids along path-2. Where R stands for H (a), CF_3 (b), Cl (c) and O-H (d).

geometrical structures are shorter than N7…H—O bond distances and are laying in between 1.319 and 1.363 Å. The C6-N7 bonds lengths in the transition states (**D**) of pathway-2 are somewhat longer than that of pathway-1 (Table 3). The energies barriers computed at MP2/6-311 + G(d,p) for TMZ-H, TMZ-CF_3, TMZ-Cl and TMZ-OH along pathway-2 are, respectively, 7.18, 4.38, 6.67 and 7.75 kcal/mol higher compare to their counterparts of pathway-1 indicating that pathway-1 is still preferred over pathway-2 in accordance with the work of Mirzaei et al. [5] which revealed that the hydrolysis of TMZ-H is more favorable to occur kinetically through the path leading to **C** (path-1) than **D** (path-2). The computed activation energy of TMZ-H along path-2 is 0.09 and 1.05 kcal/mol lower than TMZ-Cl and TMZ-CF_3; these barrier energies differences are smaller and reside within the margin of error of MP2 height barrier calculations [27,28].

The intrinsic reaction coordinate (IRC) method was computed at M062X/6-31G(d) to connect the transition states and the reactants (TMZ-R + H_2O) and products (MTIC acids) of TMZ-R (Fig. 4). The IRC calculations are also consistent with the activations energies showing the transitions states along pathway-1 to lie lower on the potential energy surface compare to that of pathway-2. The MTIC acids obtained from IRC calculations are local minima on the potential energy surfaces of TMZ-R and undergo a conversion toward global minima. We have performed the conformational analysis to find the most stable MTIC acids.

3.4. Conformational analysis of MTIC acids

Different conformers of MTIC acid can be obtained from the ring opening of TMZ-H. In order to identify different conformers of **C** and **D**, the potential energy surfaces (PES) were constructed by changing simultaneously two torsion angles (C1-N9-N8-N7 and C1-N5-C6-O11) of **C** and **D** geometries (see Fig. 5).

A careful analysis of the potential energy surfaces yields 14 conformers. The optimized geometries and relative energies of those conformers computed at MP2/6-311 + G(d,p) are given in Fig. 6. Detailed information on their geometric parameters are provided in Table 4.

These conformers differ by the values of the C1-N9-N8-N7 and C1-N5-C6-O11 torsion angles and IHBs (N9…H—O, N8…H—O15, and N9—H…O11 bonds). Conformers **1–2** are identified as the lowest in water solvent following by **3** and **4** lying 0.28 kcal/mol above **1** and **2** at MP2/6-311 + G(d,p). **1** is characterized by N9—H…O11 of 1.776 Å, and C1-N9-N8-N7 and C1-N5-C6-O11 torsion angles of 25.56 and 177.57 degrees. **2**, **3** and **4** conformers do not have intramolecular hydrogen bonds and differ with **1** by less than 0.3 kcal/mol in water solvent. Intramolecular hydrogen bonds do not increase stability of conformers in water solvent as illustrated by the relative energy of conformer **11** estimated about 5.61 kcal/mol with respect to **1**. **D** conformers are more stable than **C** conformers. The ring opening of TMZ-H along pathway-1 yields to the lowest **C** conformer identified as **7** laying 3.07 kcal/mol above **1**. The 5-(3-methyltriazen-1)imidazole-4-carboxamide acid (MTIC acid) reported by Mirzaei [5] corresponds to the open chain **9** and lies 1.09 kcal/mol above **7** in water solvent. The geometrical parameters of **1** and that of their counterparts substituted

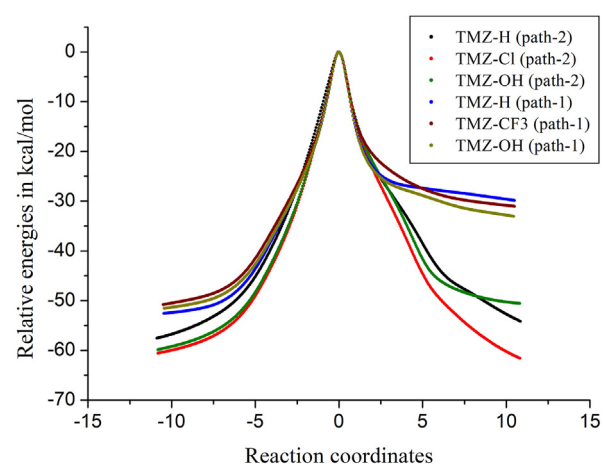


Fig. 4. Intrinsic reaction coordinate (IRC) of the formation of substituted MTIC acids obtained at M062X/6-31G(d) in water. Transitions states are considered as reference.

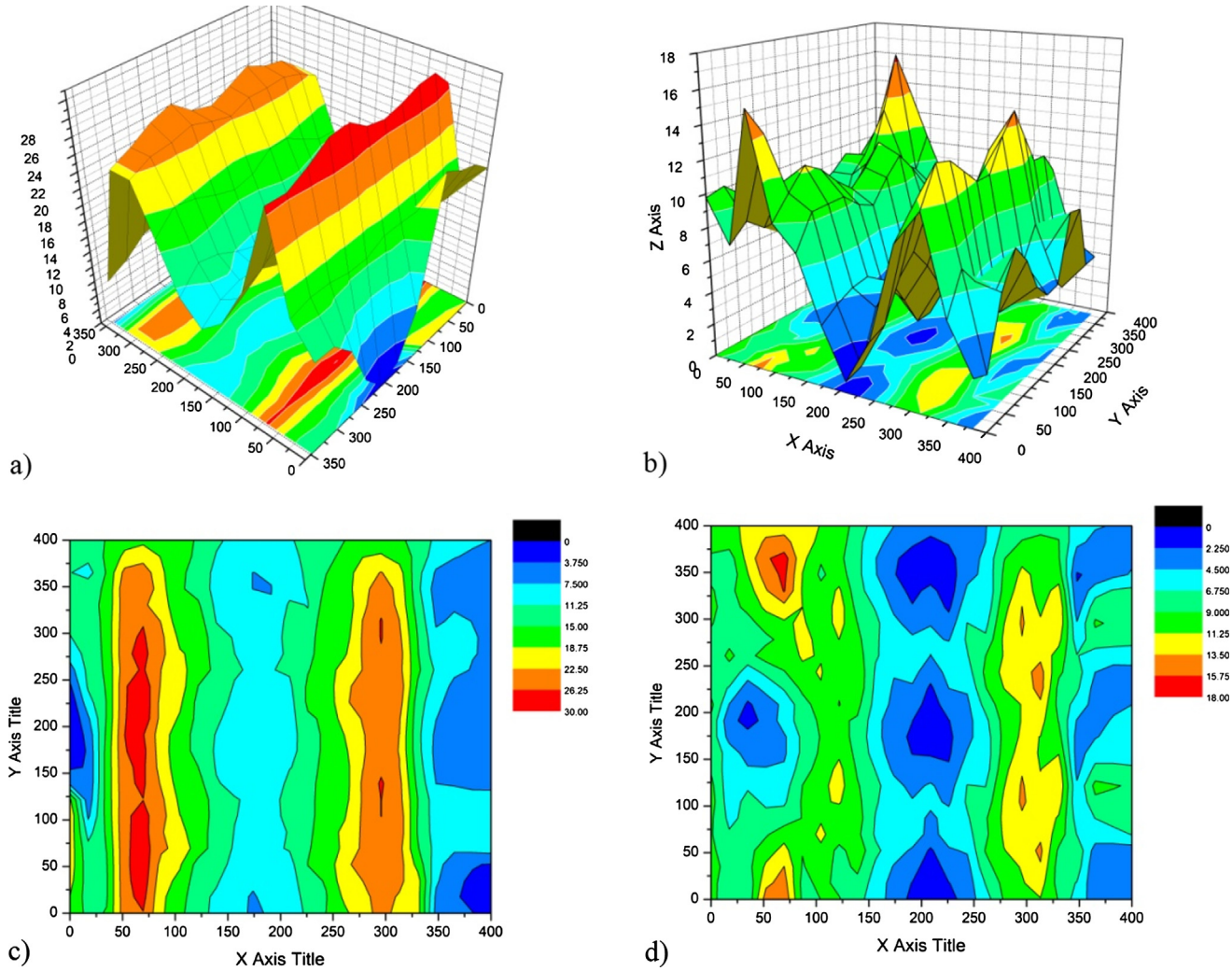


Fig. 5. Potential energy surface of TMZ-H + H_2O open ring (**C** left and **D** right) computed at B3LYP/6-31 + G(d,p). X and Y stand for C1-N9-N8-N7 and C1-N5-C6-O11 angles and Z relative energies in kcal/mol.

Table 4
Selected H-bonds and torsion angles C1-N9-N8-N7 (α) and C1-N5-C6-O11 (β), in water solvent. Refer to Scheme 2 for atom labelling.

Conf.	N9…H−O	N8…H−O15	N9−H…O11	α	β
1	–	–	1.776	25.56	177.57
2	–	–	–	20.59	−21.18
3	–	–	–	–22.00	−150.64
4	–	–	–	−157.61	−19.76
5	1.767	–	–	−150.95	179.18
6	–	–	–	−157.55	154.51
7	1.649	–	–	1.71	−175.31
8	–	–	–	22.98	−35.95
9	–	–	–	−3.94	−15.30
10	–	–	–	−3.17	166.20
11	1.581	1.585	–	−177.65	−174.24
12	–	–	–	3.194	−28.71
13	–	1.506	–	179.76	20.55
14	–	1.499	–	179.78	−161.63

MTIC acids, their HOMO-LUMO gaps, their formation energies arising from the ring opening reaction computed as total electronic energies difference between substituted MTIC acids and the molecular system TMZ-R + H_2O computed at MP2/6-311 + G(d,p) are provided in Table 5.

The data as listed in the Table shows that the HOMO-LUMO gap energies (H-L = 11.46 eV) and the formation energies (FE = 8.57–10.79 kcal/mol) obtained for substituted MTIC acids are somewhat approaching that of MTIC acid (H-L = 11.42 eV and FE = 9.21 kcal/mol) resulting from ring opening of TMZ-H. The formation energies of substituted MTIC acids from TMZ-CF_3, TMZ-Cl, TMZ-OH differ with that of TMZ-H by 1.58, −0.64, and 0.44 kcal/mol. Thus, the substitution of H by OH, CF_3 and Cl in TMZ-H has a little stabilizing effect on the ring opening of MTZ-R. This means that N7-chloroethyl, hydroxyethyl and trifluoroethyl analogues are expected to initiate the hydrolysis reaction as TMZ-H does. However the energy barriers (Fig. 4) are slightly higher for the ring opening mechanism of the substituted temozolomide compared to TMZ-H. The difference between NBO charges at C6 atom for different TMZ-R is small that can also justify the small difference in reaction energies of the TMZ-H + H_2O and TMZ-R + H_2O molecular systems.

4. Conclusion

MP2/6-311 + G(d,p) was used to investigate the effect of Cl, OH, and CF_3 substituent on ring opening of temolozomide upon water attack. The ring opening reactions of TMZ-R along pathway-1 and pathway-2 are found more favorable to occur kinetically and thermodynamically, respectively. The reaction along pathway-2 yields to the lowest 5-(3-methyltriazen-1)imidazole-4-carboxamide acid (**D** MTIC acid) lying about 3 kcal/mol below **C** conformer from pathway-1. The energies barriers computed at MP2/6-311 + G(d,p) for TMZ-R along

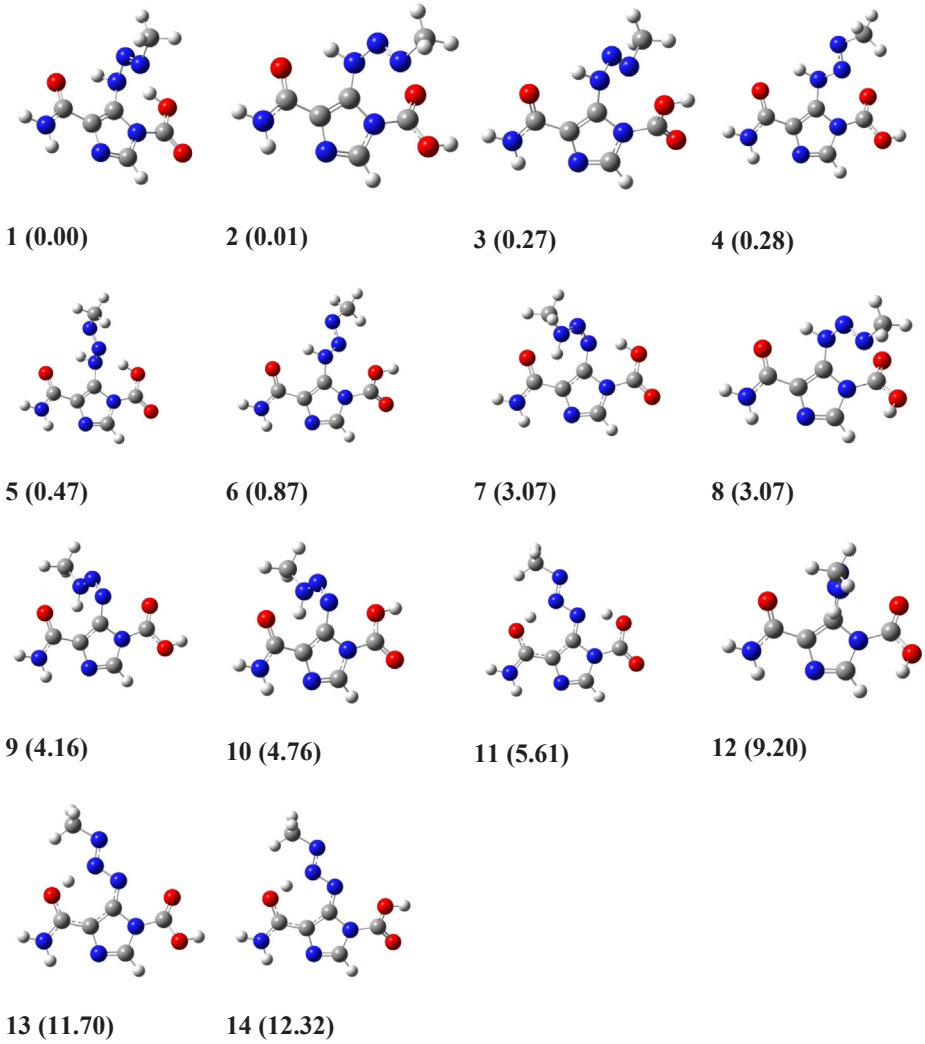


Fig. 6. Different conformers obtained after changing simultaneously C1-N9-N8-N7 and C1-N5-C6-O11 torsion angles of **C** and **D** and their relative energies in kcal/mol in parenthesis computed at MP2/6-311 + G(d,p) in water solvent.

Table 5
H-bonds, torsion angles, HOMO-LUMO gaps (in eV) and formation energies (FE in kcal/mol) of substituted MTIC acids.

MTIC	N9…O–H	α	β	H-L	FE
H	1.776	25.56	177.57	11.42	9.21
CF_3	1.807	25.02	177.04	11.46	10.79
Cl	1.807	25.05	177.32	11.46	8.57
OH	1.811	27.00	175.26	11.46	9.65
Mirzaei [5]	–	3.93	15.30	10.94	13.38

pathway-2 are 4.38–7.75 kcal/mol higher compare to pathway-1. The formation energies differences between substituted and non-substituted telemozomide are in between 0.44 and 1.58 kcal/mol. The activation energies of MTZ-CF_3, TMZ-Cl, and TMZ-OH are 1.60–3.75 kcal/mol larger than that of TMZ-H estimated about 54.21 kcal/mol. The conformational analysis reveals that intramolecular hydrogen bond does not have a stabilizing effect in water solvent and may disrupt in favor of solute-solvent intermolecular hydrogen bonds.

Declaration of Competing Interest

The authors declare that there are no known conflicts of interest.

Acknowledgments

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